MATH 110AH: Week 1 (September 25 – October 2, 2026)

Notes from the first week of MATH 110AH (Algebra, honors): magmas and the hierarchy of structures leading to groups, inverses and order, the theorem that the order of an element divides the order of the group, groups of small size, subgroups and the subgroup criterion, homomorphisms with their kernels and images, normal subgroups, quotients G/K, and the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$. Lectures and discussion sections are kept separate and ordered by date.

1 Lecture, September 25: Magmas

1.1 Course Logistics

Office hours: Friday, 4 PM. Evaluation: HW + Midterm + Final; HW is due every Monday in class (weekly).

1.2 Magmas

Definition 1.1. A "magma" M is a set with a binary operation

$$* : M \times M \to M \tag{1}$$

$$* : (m_1, m_2) \mapsto m \in M \tag{2}$$

Notation: for $*$ in M, we may write $m_1 * m_2$ instead of $*(m_1, m_2)$.

Example 1.2. $(\mathbb{R}, +)$, with $+ : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, $(\alpha, \beta) \mapsto +(\alpha, \beta) = \alpha + \beta$.

Definition 1.3. We say that a set S is closed under the operation $*$ if for every $\alpha, \beta \in S$, $\alpha * \beta \in S$.

Remark. A magma M is simply then a set with an operation such that M is closed under the operation.

Example 1.4. $([0, 1], +)$ is not a magma.

1.3 Four Definitions

Definition 1.5. A magma M is said to have a "unit" if there exists an element $i \in M$ such that $i * m = m * i = m$ for any $m \in M$.

Definition 1.6. A magma is "associative" if $a * (b * c) = (a * b) * c$ for any $a, b, c \in M$.

Definition 1.7. A magma M is "divisible" if for any $a, b \in M$, there exist $x, y \in M$ such that $a * x = b$ and $y * a = b$.

Definition 1.8. A magma M is "commutative" if for any $a, b \in M$ we have $a * b = b * a$.

1.4 The Hierarchy of Magmas

The structures: magmas, quasigroups, unital magmas, semigroups, loops, inverse semigroups, monoids, groups. The diagram from lecture, with each arrow as drawn:

HW: Complete the entire graph for EC.

Definition 1.9. A group is a monoid that has a unit, and satisfies associativity and divisibility.

From	To
Magma	Quasigroup, Unital magma, Semigroup
Quasigroup	Loop, Inverse semigroup
Unital magma	Loop, Monoid
Semigroup	Inverse semigroup, Monoid
Loop, Inverse semigroup, Monoid	Groups

Table 1: *The arrows of the magma diagram.* $(\mathbb{Q}_{>0}, \div)$ *is marked as an element of Quasigroup.*

1.5 A Quasigroup That Is Not a Loop

Example 1.10. $(\mathbb{Q}_{>0}, \div)$ is a quasigroup but not a loop. Let $a, b \in \mathbb{Q}_{>0}$. We want to find $x, y \in \mathbb{Q}_{>0}$ with $\frac{a}{x} = b$ and $\frac{y}{a} = b$, which has the solution

$$x = \frac{a}{b} \in \mathbb{Q}_{>0}, \qquad y = ab \in \mathbb{Q}_{>0}. \tag{3}$$

For a unit, we want to find $i \in \mathbb{Q}_{>0}$ such that $\frac{i}{m} = m$ for every $m \in \mathbb{Q}_{>0}$ and $\frac{m}{i} = m$ for every $m \in \mathbb{Q}_{>0}$.

Let's check associativity. Let $a, b, c \in \mathbb{Q}_{>0}$ and compute $a * (b * c)$ and $(a * b) * c$:

$$a * (b * c) = \frac{a}{b * c} = \frac{a}{b/c} = \frac{ac}{b} \tag{4}$$

$$(a * b) * c = \frac{a * b}{c} = \frac{a/b}{c} = \frac{a}{bc} \tag{5}$$

hence $a * (b * c) \neq (a * b) * c$.

1.6 Small Magmas

How to construct "small" magmas? Start with a finite set $F = \{a, b, c\}$ and fill the table of operations.

$*$	a	b	c
a	c	b	c
b	c	a	b
c	b	c	a

Table 2: *A magma on* $\{a, b, c\}$.

$$c * a = b, \quad a * b = b \quad \Rightarrow \quad \text{no unit } \times \tag{6}$$

$$\text{divisible } \checkmark \tag{7}$$

Then, for $a * b * c$,

$$a * (b * c) = a * b = b \tag{8}$$

$$(a * b) * c = b * c = b \tag{9}$$

$$(b * a) * c = c * c = a \tag{10}$$

$$b * (a * c) = b * b = b \tag{11}$$

1.7 Groups and First Examples

A group G is a magma that admits a unit, is associative, and satisfies divisibility.

Example 1.11. $(\mathbb{Z}, +)$, and $(\mathbb{Z}/4\mathbb{Z}, +) = \{0, 1, 2, 3\}$ with $[\alpha]_4 + [\beta]_4 = [\alpha + \beta]_4$.

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table 3: *Addition in* $\mathbb{Z}/4\mathbb{Z}$.

Example 1.12. Symmetries of a rectangle $R \subseteq \mathbb{R}^2$:

1. α: rotation by 180°,
2. β: reflection (horizontal),
3. γ: reflection (vertical),
4. 1: do nothing.

$\circ$	1	α	β	γ
1	1	α	β	γ
α	α	1	γ	β
β	β	γ	1	α
γ	γ	β	α	1

Table 4: *Composition of the symmetries of a rectangle.*

1.8 Uniqueness of the Identity

Proposition 1.13. *Let $(M, *)$ be a unital magma; then the identity is unique.*

Proof. Let $i, j \in M$ be two elements that satisfy the identity property. Then

$$i = i * j = j \tag{12}$$

hence $i = j$. □

1.9 Inverses

Definition 1.14. Let $(M, *)$ be a unital magma with identity $i \in M$. Let $m \in M$.

1. A "left inverse" in M is an element $n_1 \in M$ such that $n_1 * m = i$.
2. A "right inverse" is $n_2 \in M$ such that $m * n_2 = i$.

3. If an element satisfies both, it is an inverse, m^{-1}.

Example 1.15. Functions $\mathbb{R} \to \mathbb{R}$ with $\circ$: let $f(x) = e^x$ and

$$g(x) = \begin{cases} \ln(x), & x > 0 \\ 0, & x \leq 0 \end{cases} \tag{13}$$

then

$$f \circ g(x) = \begin{cases} x, & x > 0 \\ 1, & x \leq 0 \end{cases} \tag{14}$$

$$g \circ f(x) = x \tag{15}$$

Proposition 1.16. *Let $(M, *)$ be a monoid and $m \in M$ be an element. If m has a left and right inverse, then they agree, so the inverse exists.*

Proof. Let's call the left and right inverses n_1 and n_2 respectively, and compute

$$n_1 * (m * n_2) = n_1 * i = n_1 \tag{16}$$

$$(n_1 * m) * n_2 = i * n_2 = n_2 \tag{17}$$

□

Proposition 1.17. *Let $(M, *)$ be a magma with unit and divisibility. Then any element $\in M$ has an inverse.*

Proof. For all $a, b \in M$ there exist $x, y \in M$ with $a * x = b$ and $y * a = b$. Let $b = i$; then for all $a \in M$, there exist $x, y \in M$ with

$$a * x = i \tag{18}$$

$$y * a = i \tag{19}$$

□

2 Lecture, September 28: Inverses and Order

2.1 Two Definitions of a Group

Common definition of a group: magma + associativity + unit + inverses (with an arrow labeled "implies" drawn between unit and inverses).

110AH definition: magma + associativity + unit + divisibility.

Proposition 2.1. *Let $(M, *)$ be a unital magma that satisfies associativity and every element has an inverse; then M satisfies the divisibility property.*

Proof. Let $\alpha, \beta \in M$ and let β^{-1} be the inverse of β. Define $x = \beta^{-1} * \alpha$ and $y := \alpha * \beta^{-1}$; then we can see $\beta * x = \alpha$ and $y * \beta = \alpha$. □

2.2 Properties of the Inverse

Proposition 2.2. *Let $(G, *)$ be a group with identity i.*

1. *$(x^{-1})^{-1} = x$ for any $x \in G$.*
2. *$i^{-1} = i$.*
3. *The inverse of any element is unique.*
4. *If $\alpha_1, \ldots, \alpha_n \in G$, then $(\alpha_1 * \cdots * \alpha_n)^{-1} = \alpha_n^{-1} * \cdots * \alpha_1^{-1}$.*

Proof. 1. Note that $x * x^{-1} = x^{-1} * x = i$ for any $x \in G$.

2. Note that $i * i = i$.

3. Let $y_1, y_2 \in G$ be two inverses of x. Then y_1 is a left inverse and y_2 is a right inverse, so $y_1 = y_2$.

4. Apply induction on n; $n = 1$ is trivial. Let's prove the statement for $n = 2$:

$$(\alpha * \beta) * (\beta^{-1} * \alpha^{-1}) = \alpha * (\beta * \beta^{-1}) * \alpha^{-1} \tag{20}$$
$$= (\alpha * i) * \alpha^{-1} \tag{21}$$
$$= \alpha * \alpha^{-1} = i \tag{22}$$

and

$$(\beta^{-1} * \alpha^{-1}) * (\alpha * \beta) = \beta^{-1} * (\alpha^{-1} * \alpha) * \beta \tag{23}$$
$$= (\beta^{-1} * i) * \beta \tag{24}$$
$$= \beta^{-1} * \beta = i \tag{25}$$

Now let's compute $(\alpha_1 * \cdots * \alpha_n)^{-1}$. Write $\beta = \alpha_1 * \cdots * \alpha_{n-1}$; then

$$(\alpha_1 * \cdots * \alpha_n)^{-1} = (\beta * \alpha_n)^{-1} = \alpha_n^{-1} * \beta^{-1} \tag{26}$$
$$= \alpha_n^{-1} * \alpha_{n-1}^{-1} * \cdots * \alpha_1^{-1} \tag{27}$$

□

2.3 Powers

Definition 2.3. Given $\alpha \in G$, $\alpha^n = \underbrace{\alpha * \cdots * \alpha}_{n}$.

Proposition 2.4.

$$(\alpha^n)^{-1} = \alpha^{-n} \tag{28}$$
$$(\alpha^n)^m = \alpha^{nm} \tag{29}$$
$$\alpha^n * \alpha^m = \alpha^{n+m} \tag{30}$$

2.4 Order

Definition 2.5. 1. The "order" of G is the number of terms in G.

2. The "order" of $a \in G$ is the smallest positive integer n with $a^n = i$. If n does not exist, we say the order of a is infinite.

Notation: $O(G)$, $O(\alpha)$.

Theorem 2.6. *Let $\alpha \in G$ be an element of the group; then $O(\alpha)$ divides $O(G)$.*

Example 2.7. $G = (GL(2, \mathbb{R}), \times)$. Then

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad O(A) = 2 \tag{31}$$

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad O(B) = \infty \tag{32}$$

Proof. Let $H = \{i, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$ where $n = O(\alpha)$; all elements are different. Consider

$$xH = \{x, x * \alpha, x * \alpha^2, \dots, x * \alpha^{n-1}\} \tag{33}$$

If $x, y \in G$, then $xH = yH$ or $xH \cap yH = \emptyset$: if $x * \alpha^{n_1} = y * \alpha^{n_2}$, then $x = y * \alpha^{n_2 - n_1}$, so

$$xH = \{x, x * \alpha, x * \alpha^2, \dots\} = \{y * \alpha^{n_2 - n_1}, y * \alpha^{n_2 - n_1 + 1}, \dots\} = yH \tag{34}$$

Hence,

$$G = \bigcup_x xH = x_1 H \sqcup \cdots \sqcup x_k H \tag{35}$$

$$O(G) = k \cdot |H| = k \cdot O(\alpha) \tag{36}$$

□

3 Discussion, September 29: Divisibility and Powers

3.1 Loops and Units

Identity + Latin square = loop.

Proposition 3.1. *Associativity + divisibility ⇒ unit.*

Proof. Divisibility: for all $a, b \in S$ there exist $x, y \in S$ such that $ax = b$ and $ya = b$. Take $b = a$: there exist $x, y \in S$ such that $ax = a$ and $ya = a$. For arbitrary $b \in S$, there exist $u, v \in S$ such that $au = b$, $va = b$. Then

$$bx = (va)x \tag{37}$$

$$= v(ax) \tag{38}$$

$$= va \tag{39}$$

$$= b \tag{40}$$

We have $b(ax) = (ba)x = ba$; varying ba gives that x is a right unit for S. A similar argument works for y. □

3.2 Powers of a Product

Recall: for G a group and $n \in \mathbb{Z}$, it is not always true that $(ab)^n = a^n b^n$.

Example 3.2. $G = S_3 = \{\text{bijections } \{1,2,3\} \to \{1,2,3\}\}$. Take

$$a = (1\,2) \quad (1 \to 2,\ 2 \to 1,\ 3 \to 3) \tag{41}$$
$$b = (2\,3) \quad (1 \to 1,\ 2 \to 3,\ 3 \to 2) \tag{42}$$

Note: $a^2 = b^2 = e$. For ab, do b first: $1 \to 2$, $2 \to 3$, $3 \to 1$, so $ab = (1\,2\,3)$ and

$$(ab)^2 = (3\,2\,1) \neq e = a^2 b^2 \tag{43}$$

Proposition 3.3. *If $a, b \in G$ commute ($ab = ba$), then for all $n \in \mathbb{Z}$, $(ab)^n = a^n b^n$.*

Proof. Cases $n = 0$, $n > 0$, $n < 0$.

$n = 0$: $e = e \cdot e = e$ ✓

$n > 0$: induction. For $n = 1$: $ab = ab$ ✓. Assume $(ab)^n = a^n b^n$, $n > 0$. Then

$$(ab)^{n+1} = (ab)(ab)^n \tag{44}$$
$$= a^n b^n (ab) \tag{45}$$
$$= a^n (b^n a) b \tag{46}$$

and (quick induction) $b^n a = a b^n$, so

$$a^n (b^n a) b = a^n (a b^n) b \tag{47}$$
$$= (a^n a)(b^n b) \tag{48}$$
$$= a^{n+1} b^{n+1} \tag{49}$$

□

Exercise: Prove for $n < 0$ (no induction).

Exercise: Show that if $(ab)^2 = a^2 b^2$ for all $a, b \in G$, then G is abelian.

HW: find 2 groups for $O(G) = 6, 8, 9$.

4 Lecture, September 30: Order and Subgroups

4.1 Order of an Element

Definition 4.1. The "order" of a group G is the number of elements.

Definition 4.2. In a unital magma M with unit i, we define the order of an element $\alpha \in M$ to be the smallest positive integer $n \in \mathbb{Z}_{\geq 1}$ for which $\alpha^n = \alpha \cdot \alpha \cdots \alpha = i \in M$. If n does not exist, α has infinite order, $O(\alpha) = \infty$.

Example 4.3. $M = (M_{2\times 2}(\mathbb{R}), \times)$, unit $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $|M| = O(M) = \infty$. Let

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix},\ n \geq 1 \tag{50}$$

then $O(A) = 2$, $O(B) = 4$, $O(C) = \infty$, $O(I_2) = 1$.

4.2 The Order of an Element Divides the Order of the Group

Theorem 4.4. *Let G be a group and $\alpha \in G$ an element; then $O(\alpha)$ divides $O(G)$.*

Proof. Let's assume G is a finite group and $\alpha \in G$ has finite order. Let $n = O(\alpha)$ and consider the set $H = \{i, \alpha, \alpha^2, \ldots, \alpha^{n-1}\}$, $|H| = n$. We will study

$$xH = \{x, x \cdot \alpha, x * \alpha^2, \ldots, x * \alpha^{n-1}\} \tag{51}$$

for various $x \in G$.

We claim that for $x, y \in G$, the sets xH and yH are either disjoint or equal. Assume that $\beta \in xH \cap yH$. Hence $\beta = x * \alpha^p$ in xH and $\beta = y * \alpha^m$ in yH, hence $x * \alpha^p = y * \alpha^m$. We multiply by α^{-m} on the right:

$$x * \alpha^p * \alpha^{-m} = y * \alpha^m * \alpha^{-m} \tag{52}$$

$$x * \alpha^{p-m} = y \tag{53}$$

We have $y = x * \alpha^{p-m}$, so

$$yH = x * \alpha^{p-m} H = \{x * \alpha^{[p-m]_n}, x * \alpha^{[p-m+1]_n}, \ldots, x * \alpha^{[p-m+n-1]_n}\} \tag{54}$$

$$= \{x * \alpha^0, x * \alpha, \ldots, x * \alpha^{n-1}\} = xH \tag{55}$$

Then, write

$$G = \bigcup_{x \in G} xH = x_1 H \sqcup \cdots \sqcup x_k H \tag{56}$$

The union over $x \in G$ may have a lot of repetition; it becomes disjoint by the fact that the xH's are equal or disjoint.

Note: xH has exactly n elements. Take $xH = \{x, x * \alpha, \ldots, x * \alpha^{n-1}\}$ and $0 \leq p < m \leq n - 1$. If $x * \alpha^p = x * \alpha^m$, multiply by x^{-1} on the left:

$$\alpha^p = \alpha^m \tag{57}$$

$$\alpha^{p-m} = i \tag{58}$$

with $0 < m - p < n$, a contradiction. Hence,

$$|G| = \sum_{i=1}^{k} |x_i H| = nk \tag{59}$$

so $n \mid |G|$. □

4.3 Groups of Small Size

1. $|G| = 1$: then $G = \{i\}$, $i * i = i$.
2. $|G| = 2$: $G = \{\alpha, \beta\}$ with $\alpha = i$, $\beta^2 = i$, so $G = \{i, \beta\}$ with $\beta * \beta = i$, $i * \beta = \beta$, $\beta * i = \beta$, $i * i = i$.
3. $|G| = 3$: $O(\alpha) = 1$ means $\alpha = i$; otherwise $O(\alpha) = 3$, and $G = \{i, \alpha, \alpha^2\}$ with $\alpha^3 = i$.
4. $|G| = 4$: for $\alpha \in G$, $\alpha \neq i$, $O(\alpha) = 2$ or 4 ($\alpha^4 = i$).
   (a) Assume $O(\alpha) = 4$: $G = \{i, \alpha, \alpha^2, \alpha^3\}$.

(b) Otherwise all the non-identity elements have order 2: $G = \{i, \alpha, \beta, \gamma\}$ with $\alpha^2 = \beta^2 = \gamma^2 = i$. Then

$$\alpha * \beta = i \Rightarrow \beta = \alpha^{-1} \Rightarrow \beta = \alpha \quad \times \tag{60}$$

$$\alpha * \beta = \alpha \Rightarrow \beta = i \quad \times \tag{61}$$

$$\alpha * \beta = \beta \Rightarrow \alpha = i \quad \times \tag{62}$$

so $\alpha * \beta = \gamma$: the product of two non-identity elements is the third non-identity element.

$*$	i	α	β	γ
i	i	α	β	γ
α	α	i	γ	β
β	β	γ	i	α
γ	γ	β	α	i

Table 5: *The group of order 4 with every non-identity element of order 2: the symmetries of a rectangle.*

4.4 Algebraic Structures

Structure	Substructure	Functions
Linear spaces	Linear subspaces	Linear transformations
Groups	Subgroups	Homomorphisms
Rings	Ideals	Ring homomorphisms
Fields	Subfields	Field extensions

Table 6: *Algebraic structures, their substructures, and their functions.*

4.5 Subgroups

Definition 4.5. Let G be a group with multiplicative structure $*$ and identity $i \in G$. A subgroup H is a subset of G closed under $*$, that contains $i \in H$, and is closed under inverses.

Proposition 4.6. *A subset $H \subseteq G$ is a subgroup if and only if $H \neq \emptyset$ and xy^{-1} is in H for all $x, y \in H$.*

Proof. If $H \leq G$, then $i \in H$, so $H \neq \emptyset$. Furthermore, if $x, y \in H$, then $y^{-1} \in H$ because H is closed under taking inverses, and $xy^{-1} \in H$ because H is closed under multiplication.

Conversely, if $H \neq \emptyset$, then it contains some $\alpha \in G$; we can take the rule and apply it to $x = y = \alpha$, so $\alpha\alpha^{-1} = i \in H$. If we apply it to $x = i$ and $y \in H$, then $x * y^{-1} = y^{-1} \in H$, so H is closed under taking inverses. If we apply it to x and y^{-1}, then we get $xy \in H$ for all $x, y \in H$. □

5 Discussion, October 1: Subgroups and Homomorphisms

5.1 Subgroups

Definition 5.1. Let G be a group, and a subset $H \subseteq G$ is a subgroup of G if H itself is a group with the same operations as G. We write $H \leq G$ to mean H is a subgroup of G.

Example 5.2. The subset of $\mathbb{Z}$ given by $2\mathbb{Z} = \{2n : n \in \mathbb{Z}\}$; so $2\mathbb{Z} \leq \mathbb{Z}$.

Question: can you construct an example such that $H \subseteq G$, H is a group, but H is not a subgroup of G?

Example 5.3. $(\mathbb{Q}, \times)$ and $\mathbb{Z} \subseteq \mathbb{Q}$: $\mathbb{Z}$ is a group but not a subgroup of $\mathbb{Q}$ because $\times \neq +$ (it is a group as $(\mathbb{Z}, +)$).

Example 5.4. $\mathbb{Z} \supseteq \{\pm 1\}$.

Proposition 5.5. *Let G be a group and $H \subseteq G$. Then H is a subgroup of G if and only if $H \neq \emptyset$ and $xy^{-1} \in H$ when $x, y \in H$.*

Proof. Assume H is a subgroup of G; then prove $H \neq \emptyset$ and $xy^{-1} \in H$.

1. $H \neq \emptyset$: because the group always contains e.
2. $xy^{-1} \in H$: trivial (multiplicative closure).

Assume $H \neq \emptyset$ and for any $x, y \in H$, $xy^{-1} \in H$; then H is a subgroup. We check

1. associative,
2. inverses,
3. identity,
4. closed.

(1) True because G is a group and the operation of G is associative, so is H.

(3) Since $H \neq \emptyset$, there is an element $x \in H$; take $x = x$, $y = x$. By hypothesis $xx^{-1} \in H$, so $xx^{-1} = e \in H$.

(2) Since $e \in H$, let $x = e$, $y = a$; then $xy^{-1} \in H$, so $a^{-1} \in H$.

(4) Multiplicatively closed: pick $a, b \in H$. Since $b^{-1} \in H$,

$$(a)(b^{-1})^{-1} \in H \tag{63}$$
$$ab \in H \tag{64}$$

□

Example 5.6. Any group G always has at least 2 subgroups: $G \leq G$ and the trivial subgroup $\{e\} \leq G$.

Example 5.7. Let $\mathbb{Z}/4\mathbb{Z} := \{0, 1, 2, 3\}$; then $H := \{0, 2\}$ is a subgroup: $2 + 2 = 0$, $-2 = 2$.

Example 5.8. Klein group: $\{1, r\}$, $\{1, s\}$, $\{1, t\}$.

5.2 Group Homomorphisms

Group homomorphism: a map preserving group structure.

Definition 5.9. Let G and K be groups. A function $\phi : G \to K$ is a group homomorphism if $\phi(xy) = \phi(x)\phi(y)$ for any $x, y \in G$.

Example 5.10. Define $\phi : \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ by $\phi(n) = n \pmod 4$. Then

$$\phi(m + n) = m + n \pmod 4 \tag{65}$$
$$= m \pmod 4 + n \pmod 4 = \phi(m) + \phi(n) \tag{66}$$

Since $e_G e_G = e_G$,

$$\phi(e_G) = \phi(e_G e_G) = \phi(e_G)\phi(e_G) \tag{67}$$

and multiplying by $\phi(e_G)^{-1}$ we get $\phi(e_G) = e_K$. Next,

$$e_K = \phi(e_G) = \phi(xx^{-1}) = \phi(x)\phi(x^{-1}) \tag{68}$$

and multiplying by $\phi(x)^{-1}$,

$$\phi(x)^{-1} = \phi(x^{-1}) \tag{69}$$

5.3 Image and Kernel

Let $\phi : G \to K$ be a group homomorphism. The kernel of ϕ is

$$\mathrm{Ker}(\phi) = \{g \in G : \phi(g) = e_K\} \tag{70}$$

and

$$\mathrm{Image}(\phi) = \{\phi(g) \in K : g \in G\} \tag{71}$$

Proposition 5.11. *Both* $\mathrm{Ker}(\phi)$ *and* $\mathrm{Image}(\phi)$ *are subgroups (of* G *and* K *respectively).*

Proof. First consider: since $\phi(e_G) = e_K$, $e_G \in \mathrm{Ker}(\phi)$. Suppose $x, y \in \mathrm{Ker}(\phi)$; then $\phi(x) = e_K$ and

$$\phi(xy^{-1}) = \phi(x)\phi(y^{-1}) = \phi(x)\phi(y)^{-1} = e_K \tag{72}$$

so $\mathrm{Ker}(\phi)$ is a subgroup.

Second, let $a = \phi(x)$, $b = \phi(y)$; then $ab^{-1} = \phi(xy^{-1})$, so $ab^{-1} \in \mathrm{Im}(\phi)$, and $\mathrm{Im}(\phi) \leq K$. □

6 Lecture, October 2: Homomorphisms, Normal Subgroups, Quotients

6.1 Homomorphisms, Kernel, and Image

Definition 6.1. A "group homomorphism" is a function $\varphi : G \to H$ between two groups such that

1. $\varphi(a *_G b) = \varphi(a) *_H \varphi(b)$ (φ respects products),
2. $\varphi(1_G) = 1_H$ (φ respects identity).

Definition 6.2. The kernel of a group homomorphism $\varphi : G \to H$ is the set $K = \{g \in G \mid \varphi(g) = 1_H\}$.

Proposition 6.3. *Let* $\varphi : G \to H$ *be a group homomorphism; then the kernel and the image of* φ *are subgroups of* G *and* H*, respectively.*

Proof. First we show that the kernel $K := \mathrm{Ker}(\varphi)$ is a subgroup. We know that $1_G \in G$ and $\varphi(1_G) = 1_H$, so $1_G \in K$. We conclude $K \neq \emptyset$. Let $a, b \in K$ and consider $ab^{-1} \in G$. We apply φ to ab^{-1}:

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b^{-1}) = \varphi(a)\varphi(b)^{-1} \tag{73}$$

$$= 1_H \cdot 1_H^{-1} = 1_H^{-1} = 1_H \tag{74}$$

so K is a subgroup.

Now we want to prove that the image is a subgroup:

$$\varphi(G) = \mathrm{Im}(\varphi) = \{h \in H \mid \text{there exists } g,\ \varphi(g) = h\} \tag{75}$$

Note that $1_H \in \mathrm{Im}(\varphi)$, because $\varphi(1_G) = 1_H$. If $a, b \in \mathrm{Im}(\varphi)$, then there exist $x, y \in G$ with $\varphi(x) = a$, $\varphi(y) = b$. We see

$$ab^{-1} = \varphi(x)\varphi(y)^{-1} = \varphi(x)\varphi(y^{-1}) = \varphi(xy^{-1}) \tag{76}$$

so $\mathrm{Im}(\varphi)$ is a subgroup. □

6.2 Conjugation and Normal Subgroups

Proposition 6.4. *Let $\varphi : G \to H$ be a group homomorphism and let $K = \mathrm{Ker}(\varphi)$. For every element $g \in G$, we have*

$$gKg^{-1} = K, \qquad gKg^{-1} = \{gkg^{-1} \mid k \in K\} \tag{77}$$

Proof. Take $k \in K$ and apply φ to gkg^{-1}, where $g \in G$:

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) \tag{78}$$

$$= \varphi(g) \cdot 1_H \cdot \varphi(g^{-1}) \tag{79}$$

$$= \varphi(g) \cdot \varphi(g)^{-1} = 1_H \tag{80}$$

Hence $gkg^{-1} \in K$ for all $g \in G$, i.e. $gKg^{-1} \subseteq K$. Applying this containment for $g^{-1} \in G$,

$$g^{-1}Kg \subseteq K \tag{81}$$

$$gg^{-1}Kg \subseteq gK \tag{82}$$

$$gg^{-1}Kgg^{-1} \subseteq gKg^{-1} \tag{83}$$

$$K \subseteq gKg^{-1} \tag{84}$$

We conclude K is invariant under "change of coordinates". □

Definition 6.5. A group $N \leq G$ is said to be "normal" if it is invariant under conjugation; this means $N = gNg^{-1}$ for all $g \in G$. Notation: $N \trianglelefteq G$.

Q: Let G be a group. What kind of subgroups of G are kernels? For which $K \leq G$ can we find $\varphi : G \to H$ with $K = \mathrm{Ker}(\varphi)$?

1. Compare: for V, W linear spaces, $V \xrightarrow{\pi_W} V/W$ has $\mathrm{Ker}(\pi_W) = W$.

Ans: A subgroup is a kernel if and only if it is normal (invariant under conjugation).
Remark (1). If G is commutative, then for any $n \in N \trianglelefteq G$ and $g \in G$, $gng^{-1} = gg^{-1}n = n$, so $gNg^{-1} = N$.
Remark (2). Let $N \trianglelefteq G$; then $gN = Ng$. This means that for all $n \in N$ there exists $n' \in N$ such that $gn = n'g$. Compare:

1. commutative: $gn = ng$,
2. normality: $gn = n'g$, with n' possibly different from n.

6.3 Monomorphisms, Epimorphisms, Isomorphisms

Definition 6.6. 1. A "monomorphism" is a group homomorphism that is injective, i.e. $\varphi(g) = \varphi(h) \Rightarrow g = h$.

2. An "epimorphism" is a group homomorphism $\varphi : G \to H$ which is onto, i.e. for all $h \in H$ there exists $g \in G$ with $\varphi(g) = h$.

3. An "isomorphism" is a group homomorphism which is a mono- and epimorphism.

6.4 Every Normal Subgroup Is a Kernel

Theorem 6.7. *Every normal subgroup is a kernel: let $K \trianglelefteq G$; there exists a group homomorphism $\varphi : G \to H$ with $\mathrm{Ker}(\varphi) = K$.*

Proof. We define an equivalence relation on G by $a \sim_K b$ (denote: a equivalent to b modulo K) whenever $aK = bK$. We denote by G/K the set of equivalence classes $[a]_K$ of $a \in G$ (the quotient set of the equivalence relation $\sim_K$). Define

$$* : G/K \times G/K \to G/K \tag{85}$$
$$([a]_K, [b]_K) \mapsto [a *_G b]_K \tag{86}$$

1. Check the product is well defined: for $a' \sim_K a$, $b' \sim_K b$, show $[a*b]_K = [a'*b']_K$. We know $aK = a'K$ and $bK = b'K$; compute

$$a'b'K = a'bK = a'Kb = aKb = Kab = abK \tag{87}$$

hence $a * b \sim_K a' * b'$.

To check in G/K: $*$ well-defined, $*$ associative, $*$ unit, $*$ inverses (the last three are HW).

We have a group homomorphism

$$\pi_K : G \to G/K, \qquad g \mapsto [g]_K \tag{88}$$

with

$$\ker(\pi_K) = \{g \in G \mid [g]_K = [1_G]_K\} \tag{89}$$
$$= \{g \in G \mid gK = K\} = K \tag{90}$$

since $gk = 1_G \Rightarrow g = k^{-1} \in K$. □

6.5 The Unit Group

Definition 6.8. For any $m \in \mathbb{Z}$, the group of units modulo m, $(\mathbb{Z}/m\mathbb{Z})^\times$, is the set

$$\{[a] \in \mathbb{Z}/m\mathbb{Z} \mid \exists [b] \in \mathbb{Z}/m\mathbb{Z} \text{ s.t. } [ab] = 1\} \tag{91}$$

equipped with multiplication mod m.

Motivation: $(\mathbb{Z}/m\mathbb{Z}, +)$ is a cyclic group of order m, but $(\mathbb{Z}/m\mathbb{Z}, \times)$ is *not* a group.

Example 6.9. $(\mathbb{Z}/8\mathbb{Z})^\times = \{[1], [3], [7], [5]\}$.

6.6 Homomorphisms Out of $\mathbb{Z}/m\mathbb{Z}$

Recall: a homomorphism $f : (\mathbb{Z}/m\mathbb{Z}, +) \to G$ has $f([0]) = e$. Let $f([1]) = g$; we have $\text{ord}([1]) = m$. Since $\text{ord}([1]) = m$, $g^m = e$, so $\text{ord}(g) \mid \text{ord}([1])$.

Is there a homomorphism $f : \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ such that $f([1]) = [1]$? No:

$$\text{ord}([1]_{\mathbb{Z}/3\mathbb{Z}}) = 3 \tag{92}$$

$$\text{ord}([1]_{\mathbb{Z}/2\mathbb{Z}}) = 2 \tag{93}$$

In general: given G, an integer $m \in \mathbb{Z}$, and $f : \mathbb{Z}/m\mathbb{Z} \to G$,

$$\text{ord}(f[1]) \mid \text{ord}([1]) = m \tag{94}$$

More generally: given groups G, H and a homomorphism $f : G \to H$, if $g \in G$ has finite order then $\text{ord}(f(g)) \mid \text{ord}(g)$.

Application: what are all isomorphisms $f : (\mathbb{Z}/m\mathbb{Z}) \to (\mathbb{Z}/m\mathbb{Z})$? A permutation $\sigma \in S_m$ gives a set bijection, but in general not a group homomorphism.